

Lecture 2

Comparing Loss Functions and Simple Linear Regression

DSC 40A, Summer 2026

Lecture 2 part 1

Comparing Loss Functions

Announcements

release time ~2:00 pm today
should start early → HW & due Tues
→ due Fridays

- Homework 1 is due to Gradescope on Monday, July 6th.
- Discussion is today, directly after lecture at 2pm here.
 - Remember that, in general, groupwork worksheets are released on Wednesday and due Thursday. *after lecture*
 - Starting next week you will need to sign up for a check off meeting with your group.
- Look at the office hours schedule [here](#) and plan to start regularly attending!
- Remember to take a look at the supplementary readings linked on the course website.
- Please participate in lecture through hand raising, chat, and/or Q&A.

Agenda for part 1

- Recap: Empirical risk minimization.
- Choosing a loss function.
 - The role of outliers.
- Center and spread.
- Towards linear regression.
- Break!

Follow-Ups and Questions from Lecture 1

- Are there guidelines to bad loss functions?
 - Kinda: if it fails to provide useful information or guides you towards predictions that don't align with the real-world goals.
 - Non-differentiable → we will explore why in more depth later on, but for now if the loss function isn't differentiable then there's no way to tell whether that point is a minimum.
 - Unbounded or exploding slope → we will be looking into this today!
- Any other lingering questions from last time?

Question 🤔

Answer in chat or Q&A

Remember, you can always ask questions.

Recap: Empirical risk minimization

Goal

We had one goal in Lecture 1: given a dataset of values from the past, **find the best constant prediction** to make.

$$y_1 = 72$$

$$y_2 = 90$$

$$y_3 = 61$$

$$y_4 = 85$$

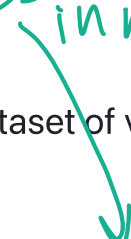
$$y_5 = 92$$

Key idea: Different definitions of "best" give us different "best predictions."

mean $\rightarrow$ median
"best" under different conditions

commute times in mins

Patterns in data



The modeling recipe

In Lecture 1, we made two full passes through our "modeling recipe."

1. Choose a model.

→ hypothesis function $h(x)$

constant $h(x) = h$ actual predicted

2. Choose a loss function.

$$L_{sq}(y_i, h) = (y_i - h)^2$$

MAE

$$L_{abs}(y_i, h) = |y_i - h|$$

⇓

$$R_{abs}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

$$h^* = \text{Median}(y_1, y_2, \dots, y_n)$$

3. Minimize average loss to find optimal model parameters.

MSE

$$R_{sq}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$
$$h^* = \text{Mean}(y_1, y_2, \dots, y_n)$$

Empirical risk minimization

- The formal name for the process of minimizing average loss is **empirical risk minimization**.

*h * optimal parameter*

constant model / parameter

- Another name for "average loss" is **empirical risk**.
- When we use the squared loss function, $L_{\text{sq}}(y_i, h) = (y_i - h)^2$, the corresponding empirical risk is mean squared error:

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$

- When we use the absolute loss function, $L_{\text{abs}}(y_i, h) = |y_i - h|$, the corresponding empirical risk is mean absolute error:

$$R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

Empirical risk minimization, in general

Key idea: If $L(y_i, h)$ is **any** loss function, the corresponding empirical risk is:

$$R(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h)$$

can find the optimal model parameter / do empirical risk minimization w/ any loss function by minimizing the average over that loss function.

Question 🤔

Answer in chat or Q&A

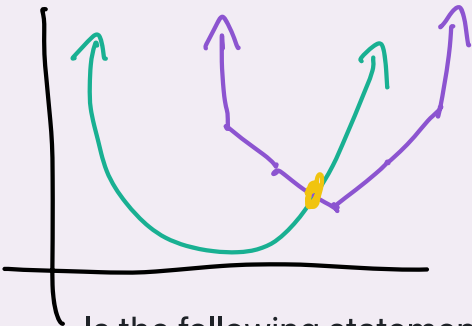
Remember, you can always ask questions.

$$(R_{\text{abs}}(h))^2 = \left(\frac{1}{n} (|y_1 - h| + |y_2 - h| + \dots + |y_n - h|) \right)^2$$

$$= \frac{1}{n^2} ((y_1 - h)^2 + |y_1 - h| \cdot |y_2 - h| + \dots)$$

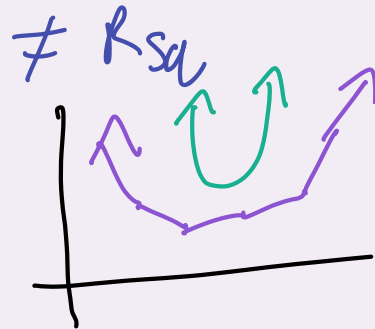
Question 🤔

Answer in chat (not Q&A)



$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$

$$R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$



Is the following statement true, for any dataset $y_1, y_2, \dots, y_n$ and prediction h ?

$$(R_{\text{abs}}(h))^2 = R_{\text{sq}}(h)$$

• A. It's true for any h and any dataset. *not true*

• B. It's can sometimes be true, but not in general.

• C. It's never true. $\rightarrow$ show is true $\geq 1 \times$

1 data point

not true

Choosing a loss function

Now what?

- We know that, for the constant model $H(x) = h$, the **mean** minimizes mean **squared** error.
- We also know that, for the constant model $H(x) = h$, the **median** minimizes mean **absolute** error.
- **How does our choice of loss function impact the resulting optimal prediction?**

$$61, 72, \boxed{85}, 90, 92 \approx \frac{400}{5} = 80$$

$$61, 72, \boxed{85}, 90, 292 \Rightarrow \frac{600}{5} = 120$$

Comparing the mean and median

- Consider our example dataset of 5 commute times.

$$y_1 = 72 \quad y_2 = 90 \quad y_3 = 61 \quad y_4 = 85 \quad y_5 = 92$$

- As of now, the **median is 85** and the **mean is 80**.

- What if we add ~~200~~ to the largest commute time, 92?

$$y_1 = 72 \quad y_2 = 90 \quad y_3 = 61 \quad y_4 = 85 \quad y_5 = \boxed{292}$$

- Now, the median is **85** but the mean is **120** !

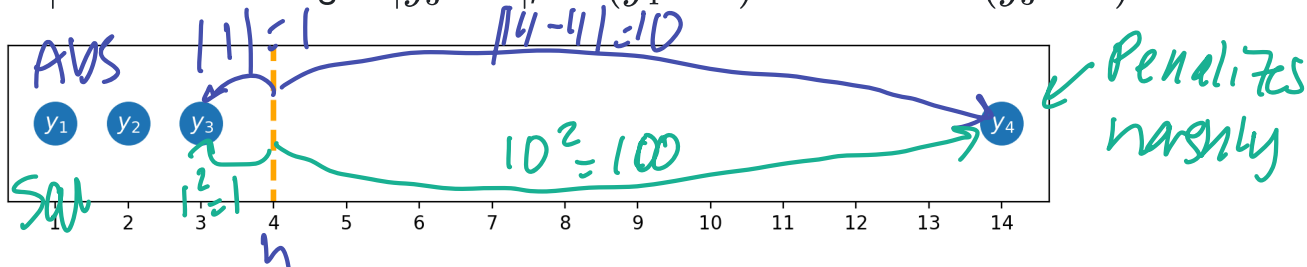
- Key idea:** The mean is quite **sensitive** to outliers.

why?

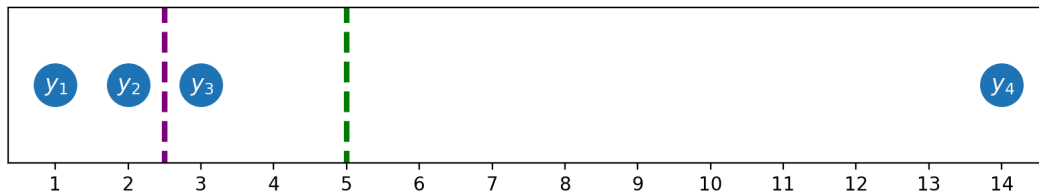
$$\begin{aligned} &\frac{200}{5} = 40 \\ &40 + 80 \\ &= 120 \end{aligned}$$

Outliers

Below, $|y_4 - h|$ is 10 times as big as $|y_3 - h|$, but $(y_4 - h)^2$ is 100 times $(y_3 - h)^2$.



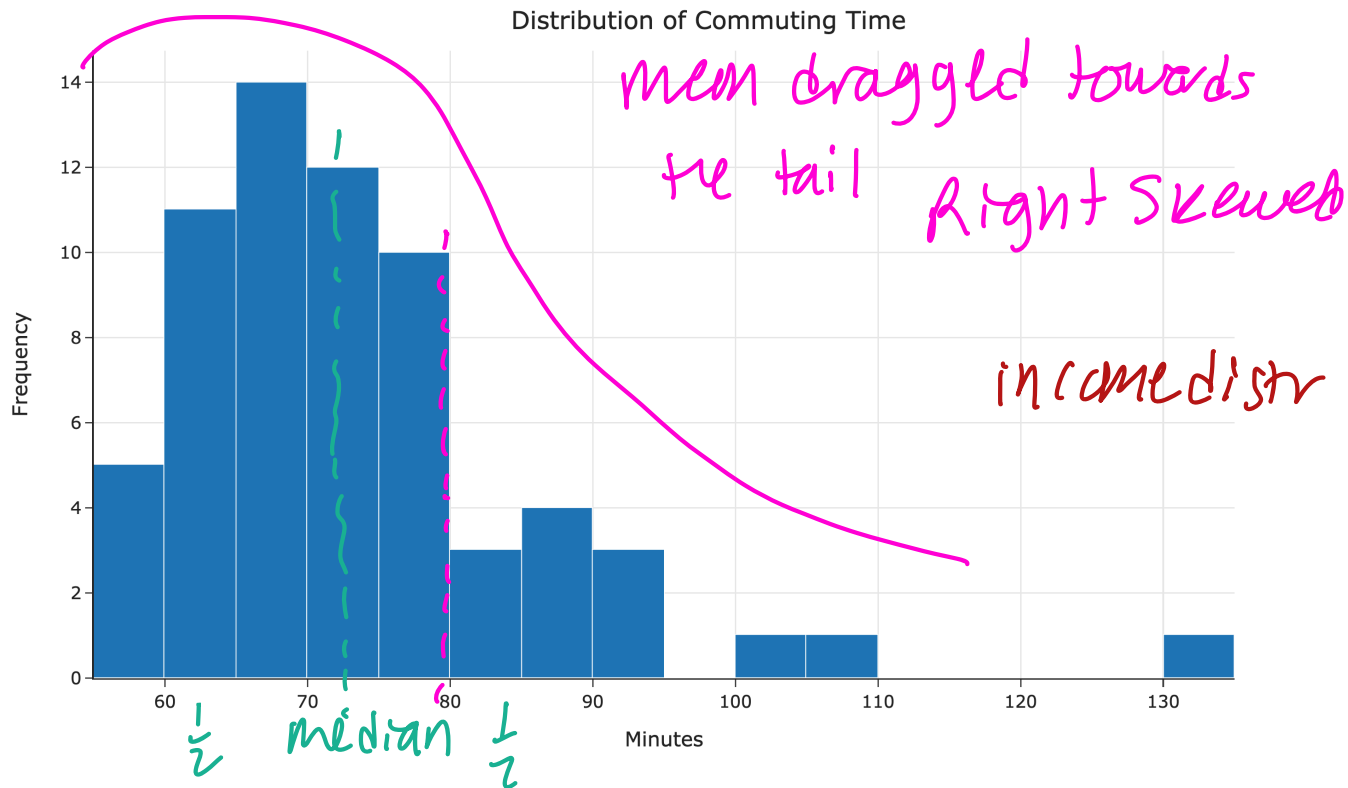
The result is that the **mean** is "pulled" in the direction of outliers, relative to the **median**.



As a result, we say the **median** is **robust** to outliers. But the **mean** was easier to solve for.

easier to minimize / easier to calculate

Real Distributions

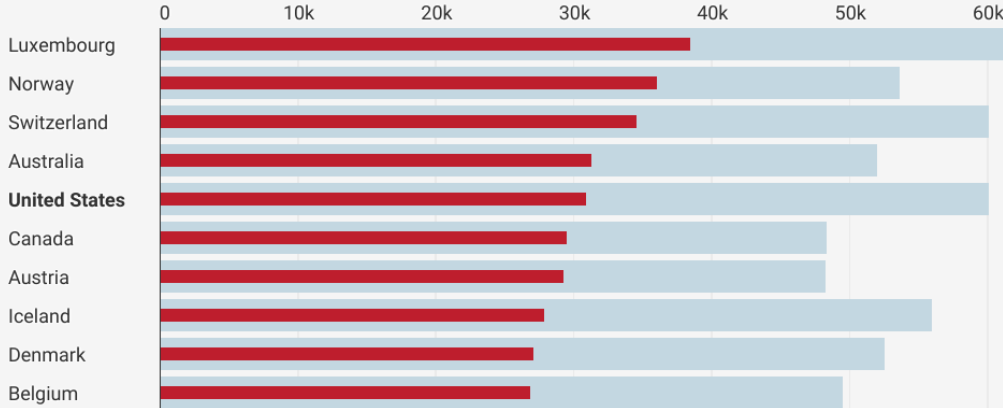


Example: Income inequality

Average vs median income

Median and mean income between 2012 and 2014 in selected OECD countries, in USD; weighted by the currencies' respective [purchasing power](#) (PPP).

■ Average income in USD ■ Median income

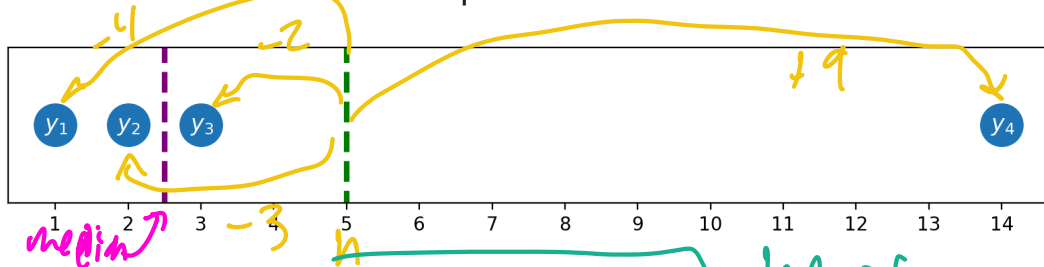


*mean influenced
by outliers*

$$\sum_{y_i < h} |y_i - h| = \sum_{y_i > h} |y_i - h|$$

Balance points

Both the **mean** and **median** are "balance points" in the distribution.



- The **mean** is the point where $\sum_{i=1}^n (y_i - h) = 0$. def of mean
 - This appears in Homework 1!

$$-1 + -3 + -2 + 9 = 0 \quad \leftarrow \text{balanced}$$

- The **median** is the point where $\#(y_i < h) = \#(y_i > h)$.

2 pts left (y_1, y_2)
2 pts right (y_3, y_4) > balanced

Why stop at squared loss?

$$\frac{1}{n} [\#(y_i < h) - \#(y_i > h)]$$

Empirical Risk, $R(h)$	Derivative of Empirical Risk, $\frac{d}{dh} R(h)$	Minimizer
$\frac{1}{n} \sum_{i=1}^n y_i - h $	$\frac{1}{n} (\sum_{y_i < h} 1 - \sum_{y_i > h} 1)$	median
$\frac{1}{n} \sum_{i=1}^n (y_i - h)^2$	$\frac{-2}{n} \sum_{i=1}^n (y_i - h)$	mean
$\frac{1}{n} \sum_{i=1}^n y_i - h ^3$		???
$\frac{1}{n} \sum_{i=1}^n (y_i - h)^4$	$-\frac{4}{n} \sum_{i=1}^n (y_i - h)^3 = 0$	???
$\frac{1}{n} \sum_{i=1}^n (y_i - h)^{100}$		???
...	...	...

take the derivative to minimize MSE →

if odd exponent w/c (-x)^3 = -LO

set = 0 find h*

diff balance point

Generalized L_p loss

For any $p \geq 1$, define the L_p loss as follows:

$$L_p(y_i, h) = |y_i - h|^p$$

The corresponding empirical risk is:

$$R_p(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|^p$$

- When $p = 1$, $h^* = \text{Median}(y_1, y_2, \dots, y_n)$. **MAE**
- When $p = 2$, $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$. **MSE**
- What about when $p = 3$?
- What about when $p \rightarrow \infty$?

Lin Alg

"p norm of vector"

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

general version of
pythagorean theorem

"squared loss in vector
form" $\rightarrow$ next week

$$\|x\|_3 = \sqrt[3]{x_1^3 + x_2^3 + \dots + x_n^3}$$

$$\|x\|_{100} = \sqrt[100]{x_1^{100} + \dots + x_n^{100}}$$

$$\|x\|_{\infty} = \max(x_1, x_2, \dots, x_n)$$

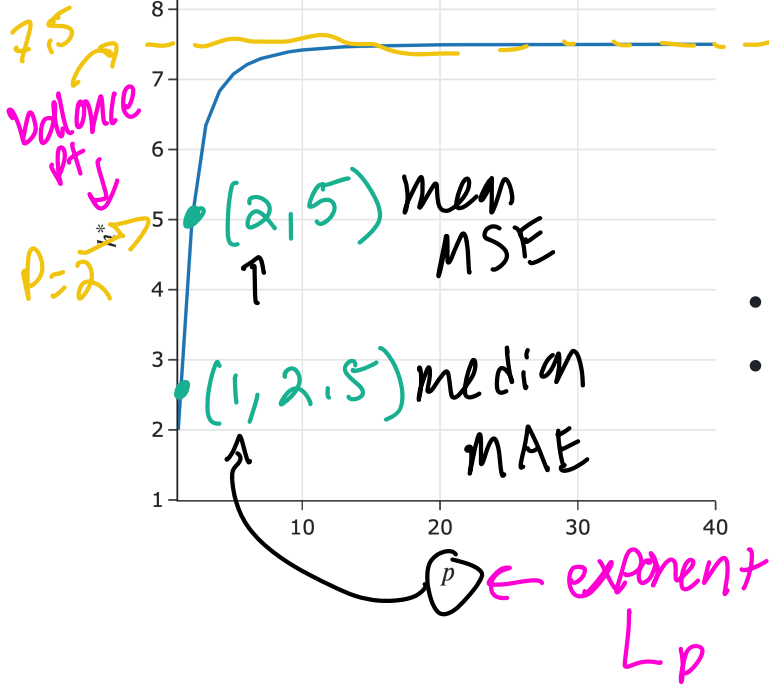
related!

relation ship

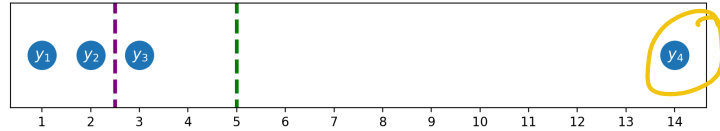
W/ L_p loss $\rightarrow$ p 's exponent
balance point / summary stat

$$7.5 = \frac{\min + \max}{2} = \frac{1 + 14}{2} = 7.5 \text{ midrange}$$

What value does h^* approach, as $p \rightarrow \infty$?



Consider the dataset 1, 2, 3, 14:



On the left:

- The x -axis is p .
- The y -axis is h^* , the optimal constant prediction for L_p loss:

$$h^* = \operatorname{argmin}_h \frac{1}{n} \sum_{i=1}^n |y_i - h|^p$$

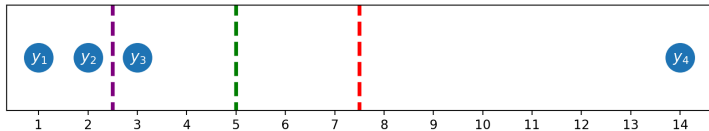
COINCIDENCE
2.5 + 5
= 7.5
happens to be
median + mean

~ "infinity loss" ~

Go back
for intuition

The **midrange** minimizes average L_∞ loss!

On the previous slide, we saw that as $p \rightarrow \infty$, the minimizer of mean L_p loss approached the midpoint of the minimum and maximum values in the dataset, or the midrange.



$\begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow 3 > 2$ but
 $3^2 > 2^2$
 $3^{100} > 2^{100}$
 $100,000!$
 $p \rightarrow \infty$

- As $p \rightarrow \infty$, $R_p(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|^p$ minimizes the "worst case" distance from any data point". (Read more [here](#)).
- If your measure of "good" is "not far from any one data point", then the midrange is the best prediction.

as $p \rightarrow \infty$
 $y^* \rightarrow 7.5$

mean = 5, worst case distance = $|14 - 5| = 9$
 median = 2.5, $|14 - 2.5| = 11.5$
 midrange = 7.5, $|14 - 7.5| = 6.5$

As you increase the exponent you increase how influenced you are by outliers

Another example: 0-1 loss

Consider, for example, the **0-1 loss**:

reward for being
close to actual
by making smaller

$$L_{0,1}(y_i, h) = \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

actual

predicted

$y_1 = 1$
 $y_2 = 2$
 $y_3 = 3$
 $h = 2$
small

The corresponding empirical risk is:

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n L_{0,1}(y_i, h)$$

avg 1s & 0s

Question 🤔

Answer in chat (not Q&A)

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

total # of pt

proportion of
pts NOT equal
to 0

Suppose $y_1, y_2, \dots, y_n$ are all unique. What is $R_{0,1}(y_1)$?

- A. 0.
- B. $\frac{1}{n}$.
- C. $\frac{n-1}{n}$.
- D. 1.

$$= \frac{\# \text{ is a } \#(y_i \neq h) + 0 \cdot \#(y_i = h)}{n}$$

$$= \frac{n-1}{n}$$

1, 2, 3, 14
what's
the mode?

Mode is not
usually unique

Minimizing empirical risk for 0-1 loss

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

Set $h^* = \text{Mode}(y_1, y_2, \dots, y_n)$ = proportion of
pts Not = h

want this to happen!
as often as possible

when is this?
 $h = ?$
mode
 $y_i = h$
as often as possible

Summary: Choosing a loss function

Key idea: Different loss functions lead to different best predictions, h^* !

Loss	Minimizer	Always Unique?	Robust to Outliers?	Differentiable?
L_{sq}	mean	yes ✓	no ✗	yes ✓
L_{abs}	median	no ✗	yes ✓	no ✗
L_{∞}	midrange	yes ✓	no ✗	no ✗
$L_{0,1}$	mode	no ✗	yes ✓	no ✗

balance ^{of} balance certain criteria

The optimal predictions, h^* , are all **summary statistics** that measure the **center** of the dataset in different ways.

1, 2, 3, 5
mean = $\frac{1+2+3+5}{4}$
= $\frac{11}{4}$
median: 2.5
2.4
2.7
|2.0000|
will minimize the MAE

Center and spread

What does it mean?

- The general form of empirical risk, for any loss function $L(y_i, h)$, is:

$$R(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h)$$

balance
pt
↓

- As we just saw, the input h^* that minimizes $R(h)$ is some measure of the **center** of the dataset.
 - Examples include the mean (L_{sq}), median (L_{abs}), and mode ($L_{0,1}$).
- The minimum output, $R(h^*)$, represents some measure of the **spread**, or **variation**, in the dataset.

Squared loss

- The empirical risk for squared loss, i.e. mean squared error, is:

$$R_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2$$

- $R_{\text{sq}}(h)$ is minimized when $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$.
- Therefore, the minimum value of $R_{\text{sq}}(h)$ is:


$$\begin{aligned} R_{\text{sq}}(h^*) &= R_{\text{sq}}(\text{Mean}(y_1, y_2, \dots, y_n)) \\ &= \frac{1}{n} \sum_{i=1}^n (y_i - \text{Mean}(y_1, y_2, \dots, y_n))^2 \end{aligned}$$

Variance

- The minimum value of $R_{sq}(h)$ is the mean squared deviation from the mean, more commonly known as the **variance**.

$$\text{Variance}(y_1, y_2, \dots, y_n) = \frac{1}{n} \sum_{i=1}^n (y_i - \text{Mean}(y_1, y_2, \dots, y_n))^2$$

Handwritten annotations:

- $\text{Mean}(y_1, y_2, \dots, y_n)$ is written in pink above the formula.
- A pink arrow points from the pink text to the mean term in the formula.
- The word "dist" is written in yellow above the minus sign, with a yellow arrow pointing to it.
- A blue bracket is drawn under the entire term $(y_i - \text{Mean}(\dots))^2$, with the text "Squared distance from the mean" written in blue below it.
- The word "Variance" is circled in blue.

- It measures the squared distance of each data point from the mean, on average.
- Its square root is called the **standard deviation**.

* $h^* = \text{mean}$

$$\underline{R_{sq}(h^*) = \text{variance}}$$

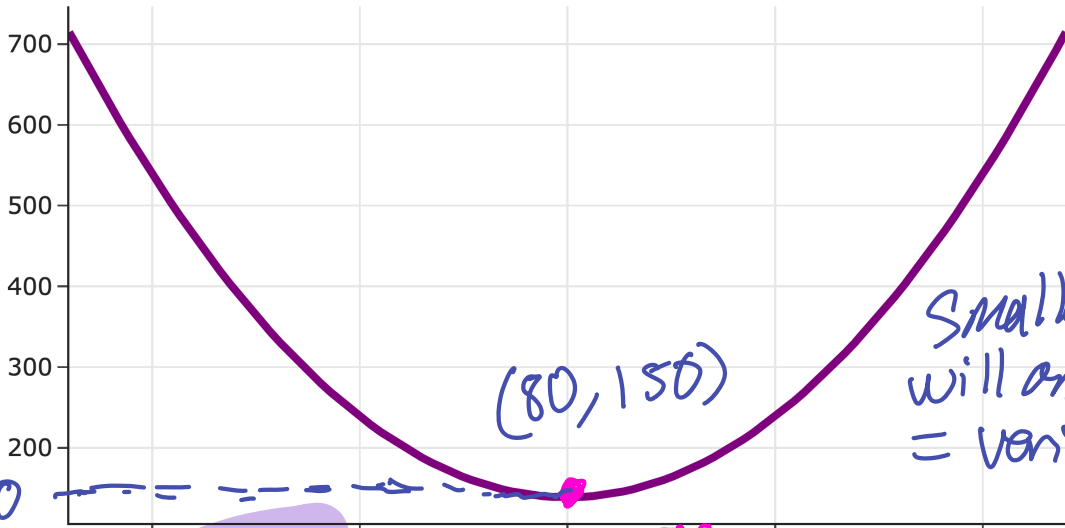
Minimizer
for MSE
is mean

Constant prediction $H(x) = h = \text{mean} = h^*$

$$R_{sq}(h) = \frac{1}{5} ((72 - h)^2 + (90 - h)^2 + (61 - h)^2 + (85 - h)^2 + (92 - h)^2)$$

MSE

$R_{sq}(h)$



Smallest MSE
will only ever
= variance

If not using the mean
the MSE WILL be higher
than the variance

$h = 80$
mean

Absolute loss

- The empirical risk for absolute loss, i.e. mean absolute error, is:

$$R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

- $R_{\text{abs}}(h)$ is minimized when $h^* = \text{Median}(y_1, y_2, \dots, y_n)$.

- ~~Therefore, the~~ minimum value of $R_{\text{abs}}(h)$ is:

$$R_{\text{abs}}(h^*) = \frac{1}{n} \sum_{i=1}^n |y_i - h|$$

$$= R_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |y_i - \text{Median}(y_1, y_2, \dots, y_n)|$$

Handwritten notes: A blue arrow points from the variable h in the second equation to the h in the first equation. Another blue arrow points from the h in the second equation to the Median term in the third equation.

search & find

MAD mean absolute deviation from the mean

'MAD from Median'

Mean absolute deviation from the median

- The minimum value of $R_{\text{abs}}(h)$ is the mean absolute deviation from the median.

$$\text{MAD from the median}(y_1, y_2, \dots, y_n) = \frac{1}{n} \sum_{i=1}^n |y_i - \text{Median}(y_1, y_2, \dots, y_n)|$$

avg actual distance to median

- It measures how far each data point is from the median, on average.

- Example:** What's the MAD from the median in the dataset 2, 3, 3, 4, 5?

median = 3

$$|2 - 3| = 1$$

$$|3 - 3| = 0$$

$$|3 - 3| = 0$$

$$|4 - 3| = 1$$

$$|5 - 3| = 2$$

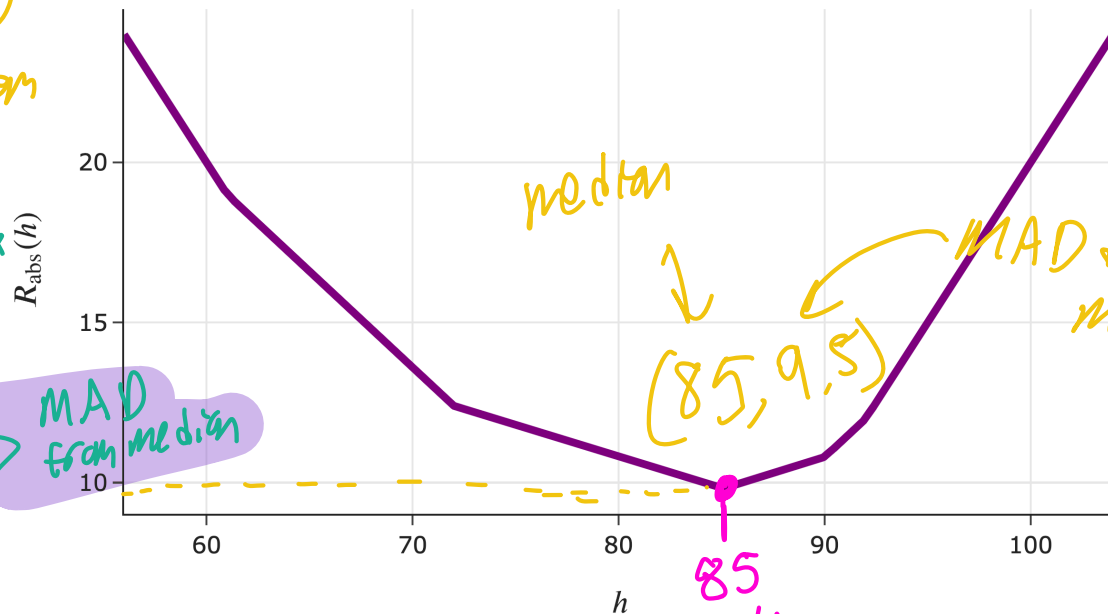
$$\frac{1 + 0 + 0 + 1 + 2}{5}$$

= $\frac{4}{5}$ MAD from the median

61 72 85 90 92

Mean absolute deviation from the median

$$R_{\text{abs}}(h) = \frac{1}{5}(|72 - h| + |90 - h| + |61 - h| + |85 - h| + |92 - h|)$$



$R_{\text{abs}}(h^*)$
 = MAD from
 median
 if $h \neq h^*$
 then

$R_{\text{abs}}(x) >$ MAD from median

0-1 loss

- The empirical risk for the 0-1 loss is:

avg over
0s & 1s

$$R_{0,1}(h) = \frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = h \\ 1 & y_i \neq h \end{cases}$$

if = then we want to set to 0
so set to 0

- This is the proportion (between 0 and 1) of data points not equal to h .
- $R_{0,1}(h)$ is minimized when $h^* = \text{Mode}(y_1, y_2, \dots, y_n)$.
- Therefore, $R_{0,1}(h^*)$ is the proportion of data points not equal to the mode.
- Example:** What's the proportion of values not equal to the mode in the dataset

2, 3, 3, 4, 5? $\frac{3}{5}$ not = 3 mode

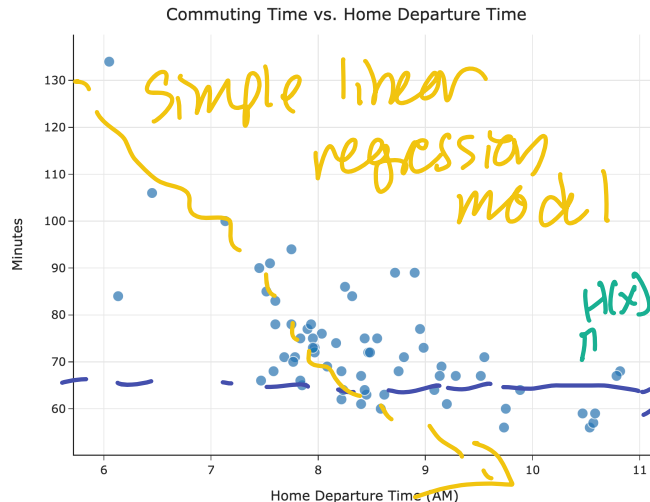
200, 3, 3, 400, 500 not great representation

Summary of center and spread

- Different loss functions $L(y_i, h)$ lead to different empirical risk functions $R(h)$, which are minimized at various measures of **center**. *→ balance pts* *avg dist*
- The minimum values of empirical risk, $R(h^*)$, are various measures of **spread**. *v/w* *aaa + bbb*
- There are many different ways to measure both center and spread; these are sometimes called **descriptive statistics**.

What's next?

Towards simple linear regression



- In Lecture 1, we introduced the idea of a hypothesis function, $H(x)$.
- We've focused on finding the best constant model, $H(x) = h$.
- Now that we understand the modeling recipe, we can apply it to find the best simple linear regression model, $H(x) = w_0 + w_1x$.
- This will allow us to make predictions that aren't all the same for every data point.

$$y = mx + b$$

↖ slope ↘ intercept

$$\nearrow H(x) = h$$

The modeling recipe

1. Choose a model.

$$H(x) = w_0 + w_1 x$$

2. Choose a loss function.

$$\rightarrow L_{sa}$$

~~L_{abs}~~

~~L_{b1}~~

need computer to calculate minimum

poor measure in
of spread
center = basic

sa b/c calculation are easy - ish(!)

3. Minimize average loss to find optimal model parameters.

$$h^*$$



$$w_0^*, w_1^*$$



$$\vec{w}^*$$

$$\begin{bmatrix} w_1^* \\ w_2^* \\ \vdots \\ w_n^* \end{bmatrix}$$

Break

$(2:4)$

Lecture 2 part 2

Simple Linear Regression

DSC 40A, Summer 2026

Part 2 Agenda

- Recap: Center and spread.
- Simple linear regression.
- Minimizing mean squared error for the simple linear model.

Question 🤔

Answer in chat; ask in Q&A

Remember, you can always ask questions!

Recap: Center and spread

avg loss $\rightarrow$ Empirical risk $\searrow$

The relationship between h^* and $R(h^*)$

- Recall, for a general loss function L and the constant model $H(x) = h$, empirical risk is of the form:

$$R(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h)$$

- h^* , the value of h that minimizes empirical risk, represents the **center** of the dataset in some way.
- $R(h^*)$, the smallest possible value of empirical risk, represents the **spread** of the dataset in some way.
- The specific center and spread depend on the choice of loss function.

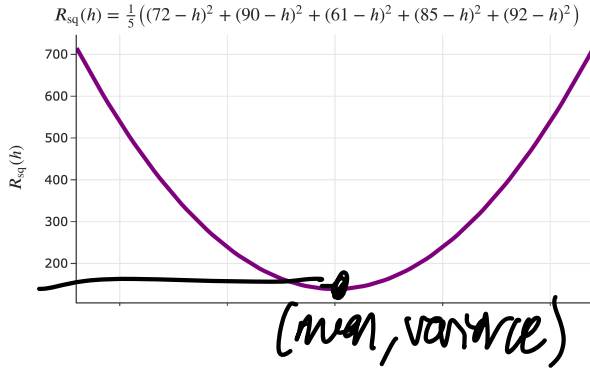
balance $\nearrow$

~~✗~~ MSE ~~✗~~ Variance

Examples

When using **squared loss**.

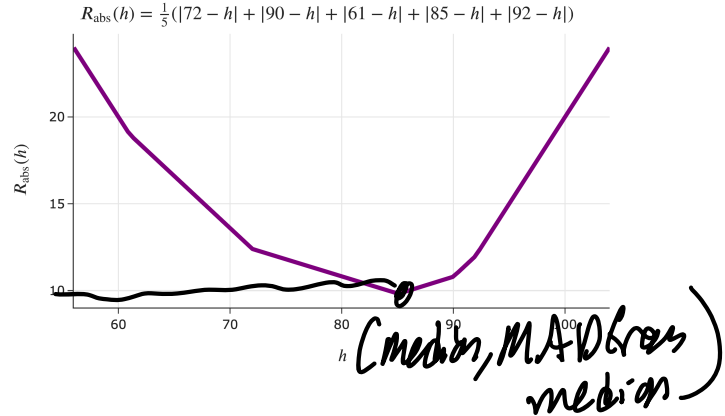
- $h^* = \text{Mean}(y_1, y_2, \dots, y_n)$.
- $R_{sq}(H^*) = \text{Variance}(y_1, y_2, \dots, y_n)$.



~~✗~~ Smaller the variance
& MAD from median are, the
more tightly clustered around the center
your data is.

When using **absolute loss**:

- $h^* = \text{Median}(y_1, y_2, \dots, y_n)$.
- $R_{abs}(h^*) = \text{MAD from the median}$.



Simple linear regression

$y = \beta_0 + \beta_1 x$ $\hat{y}$ predicted value $\hat{y}$ cost

H (time in the morning) $\rightarrow$ common time
 you leave

constant
 $H(x) = h$ $\leftarrow$ parameter
 does not use x

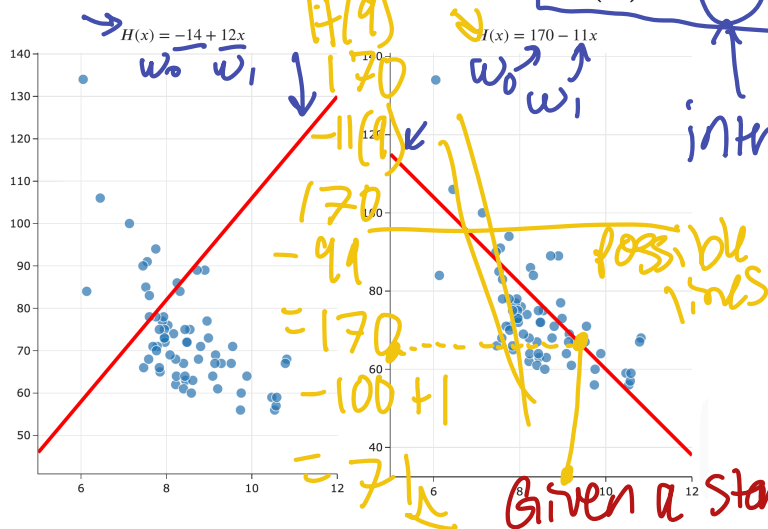
Recap: Hypothesis functions and parameters

A hypothesis function, H , takes in an x as input and returns a predicted y .

Parameters define the relationship between the input and output of a hypothesis function.

w_0 "w naught" =

The simple linear regression model $H(x) = w_0 + w_1x$ has two parameters: w_0 and w_1 .



Goal: find the BEST
 Slope (w_1^*) and BEST
 Intercept (w_0^*)

Given a start time of 8 am, it would
 take 71 mins

take 91 mins according to this Hyp function

Simple linear regression model

Now; $H(x) = w_0 + w_1 x$

The modeling recipe

1. Choose a model.

Before: $H(x) = 1$

2. Choose a loss function.

$$L_{sq}(y_i, H(x_i)) = (\overset{\text{actual}}{\downarrow} y_i - \overset{\text{predicted}}{\downarrow} H(x_i))^2$$

$$L_{abs}(y_i, H(x_i)) = |y_i - H(x_i)|$$

3. Minimize average loss to find optimal model parameters.

$$R_{sq}(H) = \frac{1}{n} \sum_{i=1}^n (y_i - H(x_i))^2$$

$$R_{abs} = \frac{1}{n} \sum_{i=1}^n |y_i - H(x_i)|$$

more common choice
+ res back to DSC 1.0

easiest to minimize ourselves

Minimizing mean squared error for the simple linear model

- We'll choose squared loss, since it's the easiest to minimize.
- Our goal, then, is to find the linear hypothesis function $H^*(x)$ that minimizes empirical risk:

$R(H)$
 $R(H=n) = R(n)$

$$R_{\text{sq}}(H) = \frac{1}{n} \sum_{i=1}^n (y_i - H(x_i))^2$$

parameters

Given

M knows
are w_0, w_1

- Since linear hypothesis functions are of the form $H(x) = w_0 + w_1x$, we can re-write R_{sq} as a function of w_0 and w_1 :

intercept *slope* *actual value*

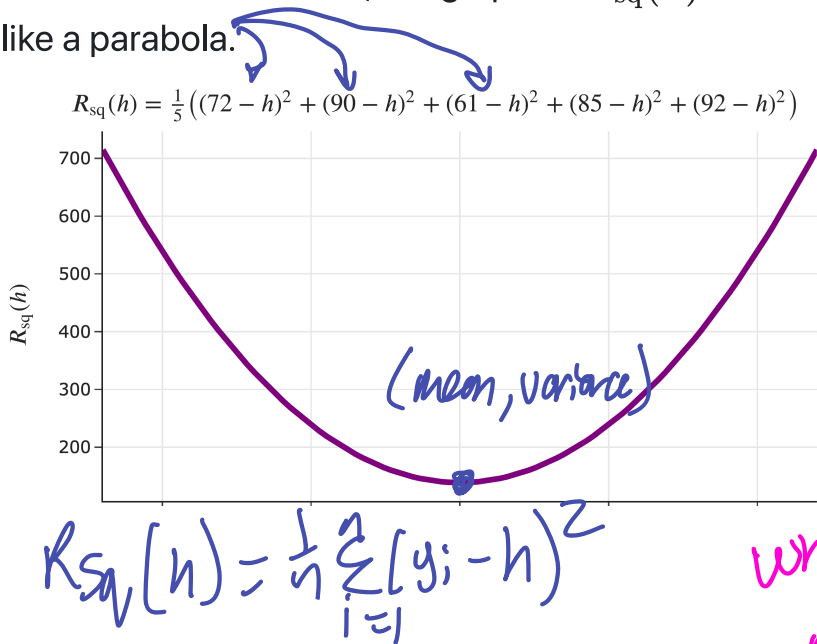
$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1x_i))^2$$

- How do we find the parameters w_0^* and w_1^* that minimize $R_{\text{sq}}(w_0, w_1)$?

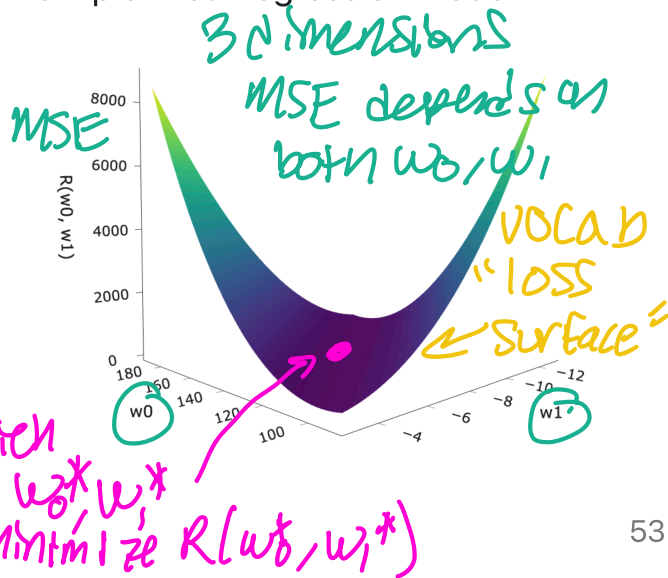
$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

Loss surface

For the constant model, the graph of $R_{sq}(h)$ looked like a parabola.



What does the graph of $R_{sq}(w_0, w_1)$ look like for the simple linear regression model?



Minimizing mean squared error for the simple linear model

Minimizing multivariate functions

- Our goal is to find the parameters w_0^* and w_1^* that minimize mean squared error:

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

- R_{sq} is a function of two variables: w_0 and w_1 .
- To minimize a function of multiple variables:
 - Take partial derivatives with respect to each variable.
 - Set all partial derivatives to 0.
 - Solve the resulting system of equations.
 - Ensure that you've found a minimum, rather than a maximum or saddle point (using the **second derivative test** for multivariate functions).

SVK
spdy

Example

$$f(x, y) = z \quad f(4, -3) = 16 - 24 + 9 - 18 = -7$$

Find the point (x, y, z) at which the following function is minimized.

$$f(x, y) = x^2 - 8x + y^2 + 6y - 7$$

Calculus

$$f_x = \frac{df}{dx} = 2x - 8 = 0$$
$$x = 4$$

$$f_y = \frac{df}{dy} = 2y + 6 = 0$$
$$y = -3$$

Completing the Square

$$(x-4)^2 - 16 + (y+3)^2 - 9 - 7$$
$$= (x-4)^2 + (y+3)^2 - 32 = 0$$

$\Rightarrow$ minimized at point $(4, -3, -32)$

x y z

Minimizing mean squared error

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

To find the w_0^* and w_1^* that minimize $R_{\text{sq}}(w_0, w_1)$, we'll:

- partial derivatives*
1. Find $\frac{\partial R_{\text{sq}}}{\partial w_0}$ and set it equal to 0.
 2. Find $\frac{\partial R_{\text{sq}}}{\partial w_1}$ and set it equal to 0.
 3. Solve the resulting system of equations.

Question 🤔

Answer in chat

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

Which of the following is equal to $\frac{\partial R_{\text{sq}}}{\partial w_0}$?

- ☒ A. $\frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$ ☐ C. $-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) x_i$
- ☒ B. $-\frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$ ☒ D. $-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

coefficient on w_0 if expanded

$$\frac{\partial R_{\text{sq}}}{\partial w_0} = \frac{1}{n} \sum_{i=1}^n 2(-1)(y_i - (w_0 + w_1 x_i))$$

$$= -\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$$

1st piece of the puzzle

*add i
for i in*

complicated

$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

dependent

$$\frac{\partial R_{sq}}{\partial w_1} = \frac{1}{n} \sum_{i=1}^n 2(y_i - (w_0 + w_1 x_i)) \underline{(x_i)} \underline{(-1)}$$

$$= -\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))$$

wrong x_i has no context outside of the sum

2nd piece of the puzzle

so

$$= -\frac{2}{n} \sum_{i=1}^n ((y_i - (w_0 + w_1 x_i)) x_i)$$

in summation

Strategy

We have a system of two equations and two unknowns (w_0 and w_1):

$$-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0$$

partial wrt w_0

To proceed, we'll:

1. Solve for w_0 in the first equation.

The result becomes w_0^* , because it's the "best intercept."

2. Plug w_0^* into the second equation and solve for w_1 .

The result becomes w_1^* , because it's the "best slope."

$$-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) x_i = 0$$

partial wrt w_1

↑
"with respect to"

$$\sum_{i=1}^n 2 = 2 + 2 + 2 \dots + 2$$

$$n \times = n \cdot 2$$

Solving for w_0^*

Goal: isolate $w_0 \Rightarrow w_0 = \dots$

$$-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0$$

$$= \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0$$

$$= \sum_{i=1}^n (y_i - w_0 - w_1 x_i) = 0$$

$$= \sum_{i=1}^n y_i - \sum_{i=1}^n w_0 - \sum_{i=1}^n w_1 x_i = 0$$

$$= \sum_{i=1}^n y_i - n w_0 - w_1 \sum_{i=1}^n x_i = 0$$

$$\frac{\sum_{i=1}^n y_i - w_1 \sum_{i=1}^n x_i}{n} = w_0$$

$$w_0 = \frac{\sum_{i=1}^n y_i - w_1 \sum_{i=1}^n x_i}{n}$$

$$w_0 = \frac{1}{n} \sum y_i - w_1 \frac{1}{n} \sum x_i$$

$$w_0^* = \bar{y} - w_1^* \bar{x}$$

Formula for the best intercept

uses the best slope

Goal: isolate $w_1^* \Rightarrow w_1 = \dots$

Solving for w_1^* ^{plug in}

$$-\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) x_i = 0$$

sometimes it's best to not multiply everything out

$$\sum (y_i - w_0 - w_1 x_i) x_i = 0$$

$$\sum (y_i - (\bar{y} - w_1^* \bar{x}) - w_1^* x_i) x_i = 0$$

$$\sum (y_i - \bar{y} + w_1^* \bar{x} - w_1^* x_i) x_i = 0$$

$$= \sum (y_i - \bar{y}) x_i + w_1^* \sum (\bar{x} - x_i) x_i = 0$$

$$\sum (y_i - \bar{y}) x_i - \underline{w_1^*} \sum (x_i - \bar{x}) x_i = 0$$

val-mean

$$w_1^* \sum (x_i - \bar{x}) x_i = \sum (y_i - \bar{y}) x_i$$

$$w_1^* = \frac{\sum (y_i - \bar{y}) x_i}{\sum (x_i - \bar{x}) x_i}$$

Least squares solutions

We've found that the values w_0^* and w_1^* that minimize R_{sq} are:

$$\Rightarrow w_1^* = \frac{\sum_{i=1}^n (y_i - \bar{y})x_i}{\sum_{i=1}^n (x_i - \bar{x})x_i} \quad \Rightarrow w_0^* = \bar{y} - w_1^* \bar{x}$$

where:

$$\begin{array}{l} \text{mean} \\ \Rightarrow \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \end{array} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

These formulas work, but let's re-write w_1^* to be a little more symmetric.

An equivalent formula for w_1^*

Claim:

$$w_1^* = \frac{\sum_{i=1}^n (y_i - \bar{y})x_i}{\sum_{i=1}^n (x_i - \bar{x})x_i} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Proof:

right numerator

$$\begin{aligned} \sum (x_i - \bar{x})(y_i - \bar{y}) &= \sum x_i (y_i - \bar{y}) - \sum \bar{x} (y_i - \bar{y}) \\ &= \sum (y_i - \bar{y})x_i - \bar{x} \sum (y_i - \bar{y}) \\ &= \sum (y_i - \bar{y})x_i \leftarrow \text{left numerator} \end{aligned}$$

Least squares solutions

re formulated

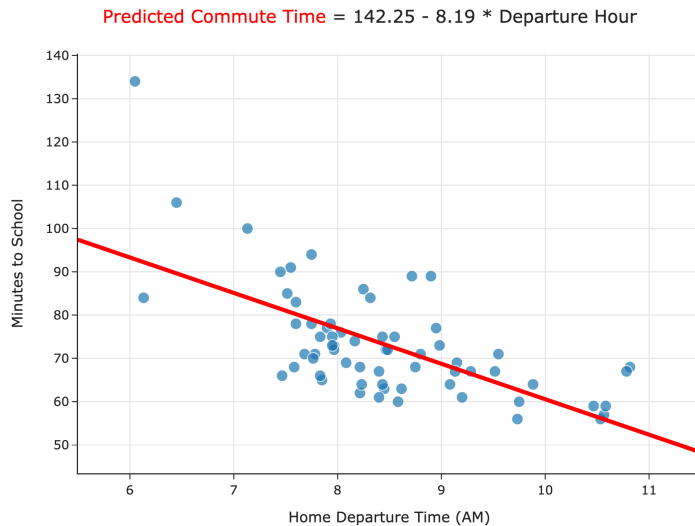
- The **least squares solutions** for the intercept w_0 and slope w_1 are:

$$w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad w_0^* = \bar{y} - w_1^* \bar{x}$$

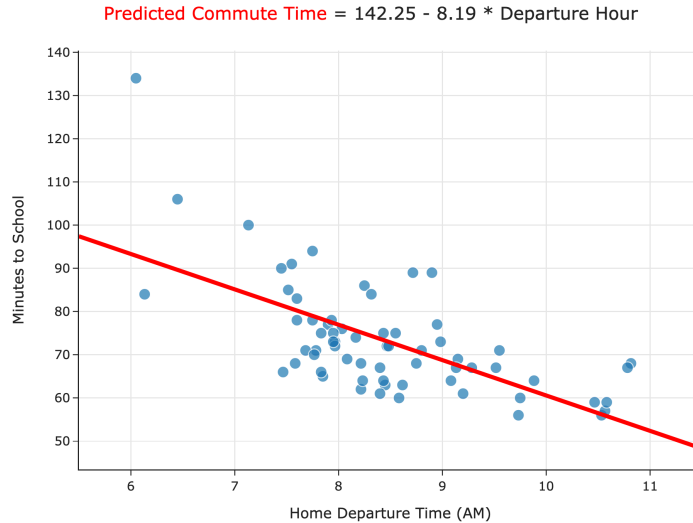
- We say w_0^* and w_1^* are **optimal parameters**, and the resulting line is called the **regression line**.
- The process of minimizing empirical risk to find optimal parameters is also called "fitting to the data." *VOLAB*
- To make predictions about the future, we use $H^*(x) = w_0^* + w_1^* x$.

Let's test these formulas out in code! Follow along by downloading the files on Piazza.

[Google Colab](#) and [commute-times.csv](#)



Causality



Can we conclude that leaving later **causes** you to get to school quicker?

What's next?

We now know how to find the optimal slope and intercept for linear hypothesis functions.

Next, we'll:

- See how the formulas we just derived connect to the formulas for the slope and intercept of the regression line we saw in DSC 10.
 - They're the same, but we need to do a bit of work to prove that.
- Learn how to interpret the slope of the regression line.
- Discuss *causality*.
- Learn how to build regression models with **multiple inputs**.
 - To do this, we'll need linear algebra!